

High-pressure processing of foods: a narrative review of lethality, injury, and quality when heat is not the kill step

Abstract

Heat remains the default way to make food microbiologically safe because it can deliver large, predictable reductions of vegetative pathogens. The same heat also drives flavour loss, enzyme damage, and protein denaturation that consumers and some product classes cannot tolerate. High-pressure processing (HPP) applies hundreds of megapascals, usually at chilled or ambient temperature, through a packed food so that pressure is uniform regardless of piece size. Vegetative cells often die; bacterial spores and several enzymes often do not. This narrative review follows those tensions as one argument rather than as a single log-reduction number. Matrix, solute, recovery medium, and injury change the kill that a given pressure appears to deliver; the same cycle may still fail legal or 5-log pasteurisation criteria in milk and juices; and quality (texture, polyphenol oxidase, bioactive proteins, allergenicity) often moves against plate counts rather than with them. Primary evidence is laboratory challenges, storage trials, one scientific opinion, and systematic or scoping reviews; almost none of it can be pooled. Those patterns do not imply that HPP is a sterilant, that it is equivalent to high-temperature short-time milk pasteurisation, or that every nonthermal field (ultrasound, plasma, irradiation) can be substituted for pressure. The next measurements that would change practice are paired pathogen and protein assays in milk, juice validations that name *Escherichia coli* O157:H7 or *Salmonella* rather than aerobic plate count, and spore kinetics recorded with in-situ temperature under pressure.

Keywords

high-pressure processing; high-pressure thermal processing; microbial inactivation; sublethal injury; milk pasteurisation; fruit juice; ready-to-eat meat

Key Summary Points

- Vegetative kill under high pressure is real, but the log reduction is not a constant read from the set-point; matrix, solute, recovery medium, and injury move it (laboratory challenges and one scoping review of ready-to-eat meats).
- Legal milk pasteurisation is not interchangeable with 400–600 MPa on the included scientific opinion.

- A 5-log drop in aerobic plate count is not a named-pathogen juice validation.
- Enzymes, bioactive proteins, and texture often move against plate counts on the same cycle, so they have to be named as endpoints.
- Spore tails remain at commercial pressures; either heat under pressure or a planned cold chain still has to carry that control.

1. Introduction

Foodborne disease is still a volume problem. The World Health Organization figures recited in fruit-and-vegetable HPP reviews put hundreds of millions of illnesses and hundreds of thousands of deaths each year on contaminated food, with vegetative bacteria, spores, parasites, and viruses as the usual agents [1]. Thermal pasteurisation and sterilisation work because time–temperature combinations can be written into law: in the European Union, milk pasteurisation is at least 72 °C for 15 s or 63 °C for 30 min, with a 5 log₁₀ performance criterion as the international reference for several milk pathogens [2]. Juice processors in the United States are told to achieve a 5-log reduction of the pertinent pathogen [3]. Those numbers are engineering specifications, not slogans. Heat meets them in most high-water-activity foods; it also cooks colour, volatiles, and whey proteins that some products are sold for.

HPP, also called high hydrostatic pressure, is the industrial attempt to keep the specification and drop the cook. Packs sit in a water-filled vessel. Pressure, typically 300–600 MPa for seconds to minutes, is transmitted isostatically: Le Chatelier’s principle, a more compact molecular order, and uniform pressure independent of geometry are the textbook triad [1,4]. Membranes leak, proteins unfold, and replication enzymes fail. Gram-negative rods are often more sensitive than Gram-positive cocci; yeasts and moulds are often easier than bacteria; bacterial spores are harder than either [1,2,5]. Commercial vessels rarely exceed about 600 MPa without an added heat supply, so surviving spores are managed by refrigeration below 7 °C rather than by a retort-like sterility [4].

The quality claim is that because the bulk temperature stays low, vitamins, colour, and “fresh-like” flavour survive better than after a heat exchanger. That claim is product-specific. Polyphenol oxidase (PPO) in grape juice can remain largely active after a pressure cycle that already cuts aerobic counts [6]. Donkey-milk lysozyme and lactoferrin lose less activity at 400 MPa than at 75 °C, but 500 MPa starts to spend that advantage [7]. Brown kelp hardness falls by tens of percent as pressure rises from 200 to 600 MPa [8]. Ready-to-eat (RTE) cooked meats are a commercial HPP success story for *Listeria monocytogenes*, yet reported log reductions at the same 500–600 MPa window still span 1–7 log colony-forming units per gram (CFU/g), and many trials used a single strain [5].

Two regulatory questions sit under the commercial success. One is whether HPP milk is equivalent to legal thermal pasteurisation. The European Food Safety Authority (EFSA) biological hazards panel modelled industrial 600 MPa for 6 min and concluded that, for the milk pathogens that could be compared, HPP delivered fewer log₁₀ reductions than the legal heat treatments [2]. Alkaline phosphatase, the heat-pasteurisation indicator, is relatively pressure-resistant and would at best flag overprocessing [2]. The other is whether juice HPP cycles that report a 5-log drop in aerobic plate count (APC) satisfy a named-pathogen HACCP plan. Several included juice papers never inoculated *E. coli* O157:H7 or *Salmonella* [3,9].

A third, quieter problem is injury. Pressure can open membrane pores that later shrink [10], hide survivors on selective agar [11], and leave cells that grow back in storage [12,13]. A plate that is blank the afternoon of treatment can still hide injured cells that grow in a recovery broth or during chilled storage. Culture-negative immediately after treatment is not the same as dead. High-pressure thermal processing (HPTP) adds heat under pressure to finish spores, at the cost of the “nonthermal” label [4,14]. Hurdles (freezing before pressure, ultrasound, plasma-activated water, plant extracts) try to buy lethality without a full cook [12,15,16,17,18,19].

The aim of this review is to compare included studies on HPP and related nonthermal processes as microbial and quality technologies, and the central argument of this review is that pressure is a reliable vegetative hurdle whose log reduction cannot be read from the set-point alone, while quality preservation is an endpoint that must be named (enzyme, protein, texture, allergen) rather than assumed.

2. Methods

2.1 Search and sources

Records came from a Scopus BibTeX export dated 20 September 2026 (359 bibliographic records, 2021–2026). Public full texts were taken only when a PDF body or Europe PMC JATS XML could be opened without a subscription. One additional plant-food HPP review was retrieved after an OpenAlex search aimed at spore kinetics [4].

2.2 Eligibility

Peer-reviewed journal articles and reviews whose titles named HPP/HHP/HPH, ultrasound, pulsed electric fields, plasma, irradiation, or nonthermal processing in a human-food, juice, meat, dairy, produce, or inactivation context. Books, chapters, conference papers, retracted items, and pet-food titles were excluded. Full-text inclusion required a readable public file of the named paper.

2.3 Study selection

After title screening, 182 records were sought. Forty-nine public files were obtained; six were excluded (wrong document, unreadable stub, bibliometric-only, off-topic production or process-control reviews). Forty-three export full texts plus one gap-fill review were included ($n = 44$). Designs range from laboratory inoculations to an EFSA opinion; they are not a single evidence grade. Screening counts live beside the manuscript, not in the Discussion. Extraction used claim-ready facts from each full text (pressure, time, organism, matrix, endpoint, and what the design cannot show). Incommensurable rows were kept in one literature table and then curated into Tables 1–3 of this article rather than averaged. Primary endpoints in those tables are named explicitly in the thematic sections that follow.

3. Pressure, membranes, and why the same megapascals do not give the same kill

Vegetative lethality under HPP is real and often large, but it is a family of outcomes, not one.

3.1 Ice changes the kill that a mild cycle can deliver

In phosphate buffer, *E. coli* K12 lost 1.83 log CFU·ml⁻¹ after 400 MPa for 9 min at 4 °C when the sample had been held at 4 °C; the same pressure–time after freezing at –24 °C or –80 °C produced 5.63 and 6.83 log reductions, with scanning electron micrographs showing wall rupture only in the frozen, pressurised cells [16]. Orange juice at 250 MPa for 15 min stayed below 0.5 log kill at pH 4.5 and 5.8 unless it was frozen first, when reductions rose to about 4 log [16]. Freezing is therefore not a footnote: it changes the physics (ice polymorphs, cell rigidity) enough to turn a mild cycle into a pasteurisation-scale kill for a surrogate organism. It also moves the process out of “HPP only.”

3.2 Strain, solute, and the ready-to-eat window

L. monocytogenes illustrates strain and solute dependence in foods that already contain *Listeria* controls. In cooked ham, lactate reduced HPP inactivation in a dose-dependent way: at 10 min, 1.4% lactate cut kill by 0.5–1.46 log versus the control depending on strain, and 2.8% lactate widened that gap [20]. That is piezo-protection: the same pressure, a more barotolerant phenotype because of an ingredient that is there for flavour and *Listeria* growth control. On RTE meats as a class, a scoping review of 74 studies found HPP in 29 of them. Below 400 MPa, inactivation was typically below 1 log even with long holds; at 500–600 MPa, reported reductions ran from 1 to 7 log CFU/g within about 5 min [5]. Thirty-one of 74 papers omitted standard deviations; 27 used a single strain; meta-analysis was judged impossible [5]. Table 1 therefore lists primary challenges as rows, not as a pooled mean.

3.3 Enumeration, injury, and repair

Enumeration can manufacture disagreement. In fruit purées treated by HPP, inactivation looked 0.9- to 4.5-fold higher in apple than in banana–apple on matched organisms [11]. Immediate reductions on non-selective media in banana–apple were 0.3 log for *Listeria*, 1.4 log for *Salmonella*, and 0.9 log for *E. coli*; acid exposure and delayed sampling changed those numbers [11]. A paper that plates only on a selective *Listeria* agar the same afternoon as treatment is not measuring the same biological residual as one that recovers injured cells. Reviews of “hidden hazardous risks” after nonthermal processing collect the same warning: culture reductions can coexist with injured survivors and later outgrowth [21].

Membrane repair is not hypothetical. After HPP, *L. monocytogenes* pore radii estimated from a fluorescence diffusion model shrank from a mean 1.338 nm on day 0 to 0.809 nm on day 4 in the least-damaged cluster, while viable counts had already fallen 7.79 log CFU/ml to below the limit of quantification and stayed there in that experiment [10]. Injury and death can travel together; they should not be collapsed into one log number copied from a graph.

Taken together, Table 1 supports a working rule: 500–600 MPa for a few minutes is the industrial vegetative window, and everything else (ice, lactate, fat, °Brix, agar, strain cocktail) is a coefficient, not noise [2,5,11,16,20]. The same rule forbids reading a 5-log claim off a pressure gauge.

Table 1. Primary HPP or HPP-hurdle microbial outcomes in the included studies (not pooled).

Paper	Matrix / organism	P/t (and extra hurdle)	Endpoint	Result
Bulut 2021 [16]	<i>E. coli</i> K12 in PBS	400 MPa / 9 min / 4 °C; ± freeze	log CFU·ml ⁻¹	1.83 (4 °C hold) vs 5.63 (–24 °C) vs 6.83 (–80 °C)
Bulut 2021 [16]	<i>E. coli</i> K12 in orange juice	250 MPa / 15 min; ± freeze	log CFU·ml ⁻¹	≤0.42 unfrozen pH 4.5–5.8; 4.15–4.88 when frozen at –80 °C
Serra-Castelló 2021 [20]	<i>L. monocytogenes</i> in cooked ham	HPP 10 min + 0–2.8% lactate	log reduction vs control	0.5–1.46 log less kill at 1.4% lactate
Torrents-Masoliver 2024 [11]	<i>Listeria</i> , <i>Salmonella</i> , <i>E. coli</i> in purées	HPP (see paper)	immediate non-selective log	banana–apple 0.3 / 1.4 / 0.9; apple higher by up to 4.5-fold
Rodríguez 2024 [15]	<i>Cronobacter sakazakii</i> (buffer)	300–400 MPa ± ultrasound	log CFU ml ⁻¹	4.0 log at 300 MPa / 10 min; no colonies at 400 MPa / 3 min or with US+HPP

Nikparvar 2021 [10]	<i>L. monocytogenes</i>	HPP (see paper)	viable count; pore radius	7.79 log to below LOQ; pores shrank over 4 days
González-Cantillo 2026 [22]	lamb burgers	long hold vs two short HHP cycles	mesophiles, <i>E. coli</i>	mesophiles 7 log to 3.7–4.5; <i>E. coli</i> to <1 log
Pimsannil 2025 [23]	precooked crab + chitooligosaccharide	100–300 MPa / 1.5 min	APC, <i>Vibrio</i>	100 MPa ~0.3–0.4 log APC; 300 MPa removed several groups
Guzel 2026 [5]	RTE meats / <i>L. monocytogenes</i>	mostly 300–600 MPa (29 studies)	reported log range	<1 log below 400 MPa; 1–7 log at 500–600 MPa

Primary endpoints in Table 1 are summarised from the notes; recovery methods differ, so rows are not interchangeable [5,11].

4. Pasteurisation performance criteria for milk and juices

A pasteurisation claim is a comparison to a legal or HACCP number, not a comparison to an untreated control.

4.1 Milk and the legal heat bar For milk, EFSA treated industrial 600 MPa for 6 min as the stringent HPP case and still found fewer log₁₀ reductions than legal heat for the pathogens that could be modelled, so HPP milk was not equivalent to pasteurisation on that metric [2]. *L. monocytogenes* data in milks show the pressure staircase: 200–250 MPa for 5–15 min often <0.5 log₁₀; 300, 350, and 400 MPa for 5 min about 1, 4.4, and 6.3 log₁₀ in cited work inside the opinion [2]. *S. aureus* is repeatedly the more resistant vegetative pathogen under pressure [2,24]. Alkaline phosphatase does not verify HPP the way it verifies heat [2].

The systematic review of raw bovine milk HPP found 65 papers with pathogen outcomes and 24 with bioactive proteins, but only four papers measured both [24]. Pressure was the strongest statistical contributor to log reductions for *L. monocytogenes*, *E. coli*, and *S. aureus* [24]. For *L. monocytogenes*, treatments above 551 MPa still needed more than about 5 min at 20–30 °C to reach >5 log in the papers that got there; come-up and come-down alone were not enough [24]. *S. aureus* kills above 5 log at 401–600 MPa often needed long holds; some >5-log *S. aureus* results used 60 °C and are no longer nonthermal [24]. That is the optimisation hole: a cycle mild enough to keep immunoglobulin G and lactoferrin is not automatically a 5-log pathogen cycle, and the experiments that would show both on the same batch are scarce [24]. A paired high-temperature short-time versus HPP bovine-milk trial that reports both endpoints on the same batches is not among the included studies.

Narrative dairy reviews still write that 400–600 MPa is “sufficient” for microbial control [25] and that ultrasound, pulsed electric fields, and supercritical carbon dioxide are promising dairy tools [26]. Those sentences compile selected rows; they do not rebut the EFSA model. Table 2 marks the difference.

Dairy experiments in this sample are consistent with “pressure helps, equivalence is not shown.” In dairy cream, 450 MPa for 5 min left total aerobic psychrotrophs similar to raw immediately and spoilage (>6 log) from day 18; 600 MPa slowed recovery but heat still produced a different trajectory [13]. In donkey milk, 200, 400, and 500 MPa cut total aerobic bacteria by 0.85, 1.58, and 4.04 log CFU ml⁻¹, while 75 °C for 2 min cut 1.34 log with much larger lysozyme loss in the results section than in the abstract [7]. Preferring the results section, 400 MPa is closer to a quality-preserving pasteurisation analogue than 500 MPa is, and neither is a pathogen challenge [7].

4.2 Juice, aerobic plate count, and named pathogens

Juices and purées look kinder to HPP because low pH already suppresses many pathogens, but the 5-log juice rule is still organism-specific. Whole Concord grape purée showed about 5-log reduction of fresh APC after both HPP and heat; yeasts and moulds were <1.0 log CFU/g after HPP versus 1.22 after heat, and HPP held those groups down longer in four-month storage [9]. That is a 5-log APC result, not a named *E. coli* O157:H7 result. Celery juices after HPP kept total aerobic counts <2 log CFU/ml for 8 weeks, with yeasts and moulds undetectable, while untreated juices spoiled in a week [27]. Semi-industrial red-globe grape juice lost about 3 log aerobic counts after HPP, ohmic heating, or thermal pasteurisation; PPO activity told a different story (HPP about 30% residual versus >90% inactivation by ohmic and thermal) [6]. A flow-through dual-frequency sonicator at 50–55 °C reached >3.6, 2.7, and 2.8 log reductions of yeast, mould, and aerobes in apple juice after 450 s efficient time and still missed a 5-log pathogen pasteurisation criterion; there was no isothermal heat-only arm [3]. None of the included juice papers reports an inoculated *E. coli* O157:H7 or *Salmonella* 5-log HPP validation.

Table 2 is therefore a scorecard of benchmarks, not of “HPP works.”

Table 2. Pasteurisation or 5-log benchmarks versus what included studies actually measured.

Paper	Product	Benchmark discussed	What was measured	Met?
Koutsoumanis 2022 [2]	ruminant milk/colostrum	legal heat pasteurisation / 5 log ₁₀ PC	modelled HPP vs heat log ₁₀	HPP 600 MPa / 6 min < heat for compared pathogens

Sykora 2026 [24]	raw bovine milk	>5 log named pathogens + native proteins	65 microbial + 24 protein papers; 4 overlap	joint optimisation not established
Agriopoulou 2023 [25]	milk products	400–600 MPa “sufficient”	narrative table of cited 5-log rows	cannot replace [2]
Machado 2023 [13]	dairy cream	spoilage delay vs heat	TAP/LAB over 51 days	450 MPa $\approx$ raw; 600 MPa slower spoilage
Koker 2023 [7]	donkey milk	TABC vs 75 °C; lysozyme/lactoferrin	factorial HHP	400 MPa $\sim$ 1.6 log TABC; not a pathogen PC
Li 2021 [9]	Concord grape purée	5-log	APC and yeasts/ moulds	5-log APC yes; named pathogens not tested
Le 2023 [27]	celery juices	refrigerated stability	TAC, yeasts/ moulds, PPO/POD	TAC held 8 weeks; not a 5-log pathogen test
Eran Nagar 2026 [6]	grape juice	industrial pasteurisation	aerobic, Y&M, PPO	$\sim$ 3 log aerobic; PPO poorly inactivated by HPP
Maghami 2026 [3]	apple juice (sonication)	FDA 5-log pathogen	native yeast/mould/ aerobes	2.7–3.6 log; 5-log not met

5. Spores, heat under pressure, and combined hurdles

Industrial HPP at ambient or chill temperature is a pasteurisation technology for vegetative cells. It is not a botulinum cook.

5.1 Spores and heat under pressure Reviews of fruit and vegetable kinetics state that pressures below 600 MPa at room temperature struggle with bacterial spores, and that some *Bacillus subtilis* spores have been reported to survive 1200 MPa at ambient temperature in laboratory equipment [1]. HPTP uses the adiabatic temperature rise (about 3 °C per 100 MPa in water-rich foods, more in fats) plus hold and decompression cooling so that time spent above 100 °C can be much shorter than a retort [14]. Commercial HPP machines usually do not supply that heat; refrigerated distribution is the practical control on spore outgrowth [4].

Houska 2022 compiles those industrial constraints with kinetic parameters rather than with a new challenge trial [4]. Kinetic form matters as much as the mean log. Vegetative inactivation in juices is often first-order, with D-values at 350 MPa on the order of 0.6–2 min for some lactic acid bacteria and yeasts [4]. Spore curves in fruit products are often concave-up (Weibull or biphasic) with tails: as time increases, remaining spores look harder [1,4]. For *Alicyclobacillus acidoterrestris* spores, a cited D-value at 600 MPa and 45 °C was 8.6 min in apple juice, 20 min in 20 °Brix lime concentrate, and 46 min in 30 °Brix blackcurrant

concentrate [4]. Soluble solids protect spores the way fat and lactate protect vegetative cells. *Neosartorya fischeri* ascospores in apple juice: 600 MPa at 75 °C for 10 min gave 3.3 log cfu/ml versus no reduction at 75 °C without pressure, in work cited by the fruit HPTP review [1]. That is synergy, and it is still not a 6-log pasteurisation of the most resistant mould spore in every concentrate.

Honey is a warning about water activity. In undiluted honey (aw about 0.55–0.57), pressure-assisted thermal processing at 600 MPa / 15 min / 85 °C and heat alone at 85 °C / 15 min both failed to reduce inoculated *B. subtilis* TEL spores, including after 24 h [28]. Diluting to aw 0.85 or 0.90 restored PATP lethality at 600 MPa / 15 min / 75 °C [28]. Low-aw foods can make “HPP plus heat” look like heat, which already failed.

5.2 Combined hurdles that are still vegetative

Hurdles other than bulk heat try to finish vegetative kill at milder set-points. A mini-review of HHP combinations in plant foods starts from pressure-resistant strains and storage regrowth after HHP alone, and treats nonthermal partners as process-design problems rather than as seasoning [12]. A hurdle review recites blueberry juice thermosonication plus 400 or 600 MPa / 5 min among combinations that let each factor run milder [19]. Ultrasound plus HPP eliminated detectable *C. sakazakii* in buffer at conditions where 300 MPa / 10 min still left about 2 log [15]. Freeze-then-pressure did the same job for *E. coli* K12 [16]. None of those combinations is a spore sterility proof.

Table 3 records spore and hurdle rows that can be cited without pretending they share a D-value.

Table 3. Spore and hurdle outcomes (reviews cited as such).

Paper	System	Process	Result
Zhang 2022 [1]	fruit/vegetable spores (review)	HPTP vs HPP vs heat	HPP ambient insufficient for many spores; example 3.3 log <i>N. fischeri</i> at 600 MPa / 75 °C / 10 min (cited)
Houska 2022 [4]	juices (review)	600 MPa ± heat	<i>A. acidoterrestris</i> D _{45°C} , 600 MPa 8.6 min apple vs 46 min 30 °Brix concentrate (cited)
Scepankova 2022 [28]	<i>B. subtilis</i> in honey	600 MPa / 85 °C / 15 min	no reduction in undiluted honey; kill after dilution to aw 0.85–0.90
Knoerzer 2025 [14]	low-acid foods (review)	HPTP vs retort	adiabatic heating shortens time above 100 °C vs >60 min retort
Xia 2022 [12]	plant foods (mini-review)		HHP alone not complete; injured-cell regrowth

		HHP + nonthermal hurdles	
Masilamani 2025 [19]	mixed foods (review)	nonthermal hurdles	combinations allow milder intensities (recited examples)
Rodríguez 2024 [15]	<i>C. sakazakii</i> buffer	US + HPP	no detectable CFU with combination
Bulut 2021 [16]	<i>E. coli</i> K12	freeze + HHP	freeze converts <2 log to 5–7 log in PBS

6. Quality endpoints that do not move with plate counts

If HPP is sold as “fresh-like,” the endpoint must be named.

6.1 Enzymes and bioactive proteins PPO and peroxidase in celery juice fell after HPP and stayed low while microbial counts held for 8 weeks [27]. In grape juice, the same class of enzyme barely moved under HPP while heat and ohmic heating removed more than 90% of PPO [6]. Cloud-stable orange juice historically uses 90–99 °C for 15–30 s to hit thermostable pectin methylesterase; some pressure studies inactivate the heat-labile PME fraction at ≥ 500 MPa, with large residual variation by cultivar [4]. Enzyme and microbe are different targets.

Proteins unfold under pressure because non-covalent bonds yield. A protein-source review reports lower allergenicity and higher digestibility/bioactivity for cereal, legume, and nut proteins after HHP, higher surface hydrophobicity for some seed and vegetable proteins, and mixed foaming/gelling/antioxidant shifts in dairy [29]. That is a functional-food story, not a Listeria story, and it is qualitative across methods. Milk bioactive proteins are the regulatory twin of section 4: holding IgG and lactoferrin is the reason to prefer HPP over Holder or high-temperature short-time in some human-milk and bovine-milk arguments, but Sykora’s four overlap papers are not enough to set a dual specification [24]. Donkey milk shows the trade in one factorial: 500 MPa maximised aerobic kill and maximised lactoferrin activity loss [7].

6.2 Texture in gels and formulated products

Texture can fail while safety succeeds. Unwashed *Saccharina latissima* hardness fell from 312 g untreated to 38 g at 600 MPa (about 88% drop), with compression distance halved [8]. That kelp is not a ready-to-eat meat, but it is a reminder that 600 MPa is not gentle for every gel. Macroalgae-fortified sausages gained minerals and, immediately after HPP, total aerobic mesophiles <1.0 log CFU/g in the fortified HPP batch [30]. Pumpkin pulp with a natural preservative plus HPP showed no microbial growth for 45 days, with better colour and firmness than a thermal comparator in the NBP formulations [31]. Minimally processed potatoes lost lightness and gained ΔE after HPP or HPP/ultrasound; Enterobacteriaceae

became undetectable after HPP, while combined ultrasound did not stay different from untreated APC by day 14 [32]. Quality and kill can trade in either direction.

6.3 Seafood storage ecology

Seafood storage trials are mostly spoilage ecology, not pathogen HACCP. Salmon and plaice fillets after HPP kept Enterobacteriaceae, mesophiles, psychrotrophs, and *Pseudomonas* lower at every sampling day and remained microbiologically acceptable to day 12, with histamine remaining low [33]. Mackerel fish balls at 250–350 MPa started <1 log CFU/g versus about 2.2 in controls and stayed down for 7 days at 4 °C, comparable to 121 °C retort on counts but not on a cooked texture [34]. Squid mantle with mint extract plus 400 MPa delayed total viable count relative to pressure or extract alone; several treatments still crossed a TVC limit by day 6 without the mint [35]. Precooked crab needed 300 MPa more than 100 MPa to move *Vibrio* and hydrogen-sulphide producers [23]. Low-temperature high-pressure carbon dioxide at 6.5 MPa / 10 °C / 15 min inactivated 99.45% of *E. coli* in suspension but only 93% on prawn surfaces (67% for *S. aureus* on prawn), so surface topography limited a process that looked excellent in buffer [36]. Lamb burgers with lupin flour dropped coliforms and *E. coli* below 1 log whether a long hold or two short HHP cycles were used [22]. Sous-vide products after high hydrostatic pressure stayed below 1.30 log CFU/g on TSA through day 45 in one trial while controls exceeded 5 log [37]. An aquatic-food UHP review collects these patterns without a pooled shelf-life [38].

Fermented and formulated drinks add a further twist: ultrasound-assisted ultra-high pressure changed antioxidant readings and sugar consumption in a lactic fermentation at 24–40 h [39]. That is process intensification of a ferment, not pasteurisation of a pathogen.

Seafood numbers in the included trials are mostly spoilage ecology. They still earn a place because RTE fish is a commercial HPP category and because histamine, *Vibrio*, and *Pseudomonas* are the practical failures, not a 5-log *Salmonella* graph. Castrica's dual-species fillet trial is the cleanest storage design: treated packs stayed acceptable to day 12 while untreated flora ran ahead, and histamine stayed low [33]. Wang's fish balls compare 250–350 MPa with a 121 °C cook on the same product; counts can look similar at day 7 while the thermal product is a different food [34]. Nilsuwan's mint-plus-pressure squid is a hurdle in the quality section rather than in section 5 because the extra kill is modest and the benefit is days of TVC delay [35]. Crab at 100 MPa barely moved APC; 300 MPa was the step that changed *Vibrio* [23]. Lian's carbon-dioxide process is not HPP, but it shows the same buffer-versus-surface gap that plasma shows on lettuce [18,36]. Pérez Alcalá's sous-vide plus pressure trial is a refrigerated RTE logic: controls crossed 5 log while pressurised packs stayed near the detection floor for weeks [37]. González-Cantillo's burger cycles (one long hold versus two shorts) both dropped *E. coli* below 1 log, so cycle architecture was not the dominant factor in

that matrix [22]. Marçal's mineral-fortified sausages and Rossi Ribeiro's pumpkin pulp show HPP as a shelf-life tool in formulated products where "fresh-like" is already compromised by other ingredients [30,31]. Peng's aquatic UHP review is the narrative envelope for those rows, not an extra dataset [38].

Taken together, HPP quality is sometimes better than heat (grape APC storage [9], donkey-milk enzymes at 400 MPa [7], avocado colour in commercial examples reviewed for plant foods [4]) and sometimes worse or simply different (kelp hardness [8], grape PPO [6], cream at 450 MPa [13]). Plate counts do not arbitrate those sentences.

7. Other nonthermal fields in the same literature

The included studies are HPP-heavy. Other fields appear as supporting technologies, not as a balanced map.

7.1 Ultrasound, plasma, and irradiation as different unit operations

Ultrasound in apple juice at reduced temperature, with dual-frequency excitation and a Venturi, reached a few log of spoilage flora and improved visual sedimentation, with a specific energy of 0.044 kWh/L at the best condition, and did not replace a 5-log pathogen pasteuriser [3]. Ultrasound reviews of mycotoxin and fungal decontamination quote high percentage degradations of aflatoxin B1 under long, high-intensity treatments that are not juice HACCP validations [40]. Sugarcane-juice comparisons of ultrasound, pulsed electric fields, HPP, and microfluidization versus heat treat HHP as the most used nonthermal option in that matrix [41].

Plasma is a surface technology. Gliding-arc plasma for 5 min plus C-phycoerythrin delayed rainbow trout total viable count crossing 6 log CFU/g from day 3 (control) to day 12, and kept total volatile nitrogen below 30 mg N/100 g through day 18, while untreated fillets did not [17]. Plasma-functionalised water on fresh-cut lettuce reached 4–5 log₁₀ laboratory reductions in multi-step washing and about 2 log₁₀ in a pilot scenario that resembled a factory wash [18]. Scale, not species, is the disagreement.

Ionising irradiation of fruit and vegetables is summarised with the FAO/IAEA/WHO statement of no toxicological problem up to 10 kGy and an FDA refrigerated-meat limit of 4.5 kGy, plus typical lethal-dose tables [42]. That is a different legal history from HPP and should not be blended into a single "nonthermal efficacy."

Broad matrix reviews rank HPP among options with typical vegetative reductions of 3–6 log₁₀ and a high technology-readiness level, while insisting that no method is universal [43]. Beverage reviews argue that nonthermal methods can match thermal preservation with better

nutrient retention, reciting a soy smoothie at 650 MPa / 20 °C / 3 min among examples [44]. Those sentences are maps. They do not override Tables 1–3. Weihe’s ranking of fields by typical log₁₀ and readiness is useful for an engineer choosing a vessel; it is not a substitute for the milk opinion or the RTE-meat scoping range [2,5,43]. Jeevitha’s smoothie example sits at 650 MPa, which is already above many commercial 600 MPa ceilings discussed for plant foods [4,44]. Hashemi’s mycotoxin percentages come from long ultrasonic treatments that would not be confused with a 3-minute HPP juice cycle if the methods were read [3,40]. Mukhtar’s sugarcane ranking of HHP versus ohmic heating is matrix-bound [41]. Zhang’s ultrasound-assisted ultra-high pressure fermentation is a different unit operation again: antioxidants and sugar use in a lactic culture, not a pathogen specification [39]. Keeping those papers in section 7 rather than folding them into Tables 1–2 avoids mixing unit operations that share a slogan more than a mechanism.

The physical picture is still pressure for HPP, cavitation for the sonicator, reactive species for plasma, and photons or electrons for irradiation. A combined factory line can use more than one, but a review that averages their log reductions would hide the scale-up failures already visible for plasma water and the 5-log miss for reduced-temperature sonication [3,18].

Discussion

Two readings of HPP can both be true. Commercially, 500–600 MPa for a few minutes is already a vegetative pasteurisation for refrigerated juices, guacamole-style herb pastes, and post-lethality RTE meats, and it avoids a second cook [4,5]. On the milk dossier the same window is not equivalent to legal pasteurisation on EFSA’s comparison: indicator enzymes do not transfer, and bioactive-protein preservation has not been co-validated with 5-log pathogen kill on enough shared batches [2,24]. Collapsing those readings into one “HPP is safe and mild” sentence is how a narrative review would mislead.

Logs cannot be pooled. Guzel’s scoping review refused a meta-analysis because variance, D-values-only papers, and single-strain challenges broke the table [5]. Torrents-Masoliver showed that fruit and agar can multiply apparent kill [11]. Serra-Castelló showed that lactate, a normal ham ingredient, subtracts kill [20]. Wu collected injury as a systematic after-effect of nonthermal culture success [21]. A reviewer who averages 1 log and 7 log at “600 MPa” is averaging different foods, strains, and methods.

Quality claims need the same discipline. PPO, lysozyme, kelp hardness, and APC are four thermometers. Eran Nagar’s grape juice is the exhibit: similar microbial drops, unlike enzyme drops, across HPP, ohmic, and thermal [6]. Braspaiboon’s allergenicity trends are a fifth thermometer still largely unconnected to the *Listeria* rows [29].

Other nonthermal fields in this sample are too thin to rank against HPP. Ultrasound missed 5-log in apple juice [3]. Plasma lost about 2–3 log moving from bench lettuce to pilot [18]. Irradiation evidence here is a mini-review [42]. High-pressure carbon dioxide looked strong in suspension and weaker on prawn [36]. Substituting “nonthermal” for “HPP” in a conclusion would erase those differences.

Low-acid vegetable juices (pH above 4.6) remain a separate safety class even when HPP has reduced vegetative counts. *Clostridium botulinum* spores can grow in that pH window if the cold chain fails; processors still have to show that the product cannot support growth and toxin within its labelled life [4]. Refrigeration after HPP is therefore not only a quality choice. It is the control that makes commercial 600 MPa cycles legal for soups and some vegetable juices that would otherwise need a full sterilisation mentality [4,14]. Fruit juices below pH 4.6 are kinder, but acid-tolerant *E. coli* O157:H7 and *Salmonella* can survive for weeks at low pH with a low infectious dose, which is why a named-pathogen 5-log test is not optional [4]. The included grape and celery papers did not run that test [9,27].

Limitations of the studies are also limitations of this review. Named-pathogen juice validations (*E. coli* O157:H7 or *Salmonella* under HPP) and a paired bovine-milk HTST versus HPP trial with both pathogen and protein endpoints are not in this sample. Abstracts can disagree with results (donkey milk [7]). Several reviews recycle the same spore examples. Honey, kelp, crab, and cream are not one population. None of the included primary papers is a randomised trial of human foodborne disease. The plant-food kinetics review adds D-values and equipment limits; it does not add a new inoculation [4].

What should be measured next follows from those holes rather than from a generic call for more work. A bovine-milk HPP versus high-temperature short-time experiment that reports the same pathogen cocktail (including *S. aureus*) and the same bioactive proteins on each batch, with recovery media stated, would close the dual-specification gap. Juice HPP validations still need inoculated *E. coli* O157:H7 or *Salmonella* against the FDA 5-log criterion rather than APC. Spore work needs measured temperature under pressure, °Brix, and a Weibull or biphasic fit instead of a single ambient-600 MPa log. RTE meat challenges that use strain cocktails, report standard deviations, and treat lactate or fat as designed factors would make the 1–7 log window interpretable [5,20]. Plasma and ultrasound still need paired bench-versus-plant measurements, given the lettuce pilot drop [18] and the sonicator’s missing heat control [3].

Conclusions

HPP reduces many vegetative bacteria in high-water-activity foods at 500–600 MPa, and that reduction is large enough, in the right matrix, to support refrigerated juices and post-lethality

meats. The reduction is not a constant, and it is not a spore kill. A cycle that looks adequate on non-selective agar the same day may look weaker after acid recovery, in a high-lactate ham, or after a week of chilled storage [11,13,20].

Legal milk pasteurisation is not interchangeable with 400–600 MPa on the included opinion: the industrial cycle underperforms heat on comparable log₁₀ reductions [2]. A 5-log APC in grape purée is likewise not a juice pathogen validation [9]. Spore tails remain, so either HPTP or a cold chain has to be planned, with water activity as a designed factor [1,4,28]. Enzymes, bioactive proteins, and texture have to be measured on the same samples as microbes, because they do not covary [6,7,8]. Recovery methods and strain sets can move a log reduction as much as 100 MPa can, and they should be reported with the set-point [5,11]. Ultrasound, plasma, and irradiation stay in separate sentences until their included primary data are as dense as the HPP rows in Tables 1–3.

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